

Camera-Based Closed-Loop Fingertip Deflection Guidance: Pilot Demonstrations in Target Acquisition and Object Retrieval

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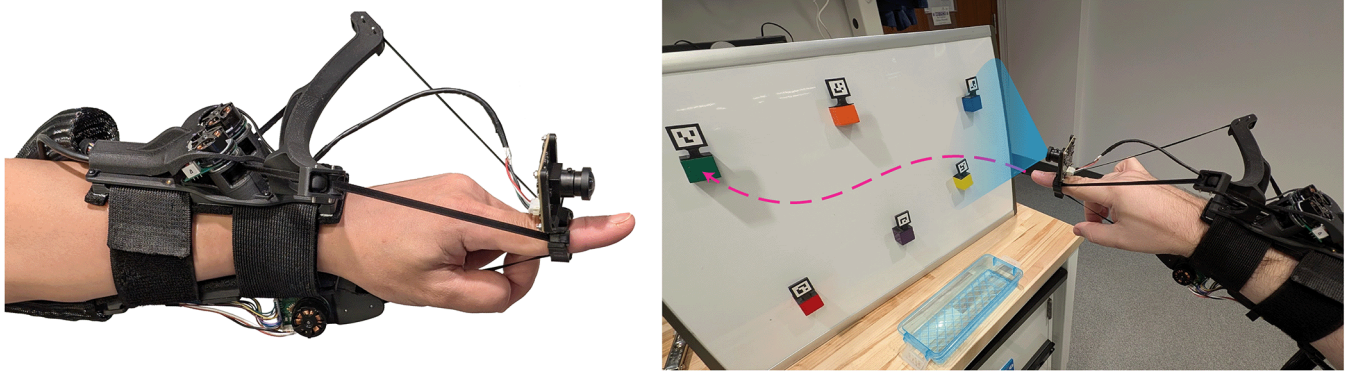


Figure 1: The NURing Device (a) as worn by the user and (b) providing continuous guidance cues during an eyes-free retrieval demonstration.

Abstract

Many eyes-free guidance systems convey direction through symbolic cues that require interpretation, potentially increasing cognitive load and competing with critical sensory channels. This work builds on prior studies of fingertip deflection as a physically-grounded guidance cue previously demonstrated in controlled 1D and 2D settings using an instrumented testbed. Here, a camera is integrated onto the actuation ring to enable object-referenced, closed-loop guidance during free reach. Using ArUco targets, the system estimates target displacement in the camera frame and maps this error into continuous fingertip deflection cues in real time. Two pilot demonstrations illustrate feasibility: (1) a Fitts-style touchscreen target acquisition task with trajectory visualization, and (2) an object-retrieval task with video-based evidence of guided approach and grasp. These vignettes ground a forthcoming study with participants who are blind and vision-impaired, and invite discussion on how ring-mounted sensing can best support embodied, eyes-free guidance in everyday interactions.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; *Accessibility systems and tools*.

Keywords

Haptics, Wearable, Accessibility, Guidance, Individuals with Vision Impairment, Assistive Technologies

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1 Introduction

For individuals with blindness or vision impairment (BVI), tasks that require locating and manipulating small objects in peripersonal space can be difficult or infeasible. Safely retrieving small items, such as dropped pills or kitchen utensils, is frequently reported as one of the most time-consuming and frustrating challenges for individuals with BVI [9] and can be safety-critical in homes shared with children or pets. Over time, repeated difficulty with everyday tasks can undermine independence and a sense of agency [1].



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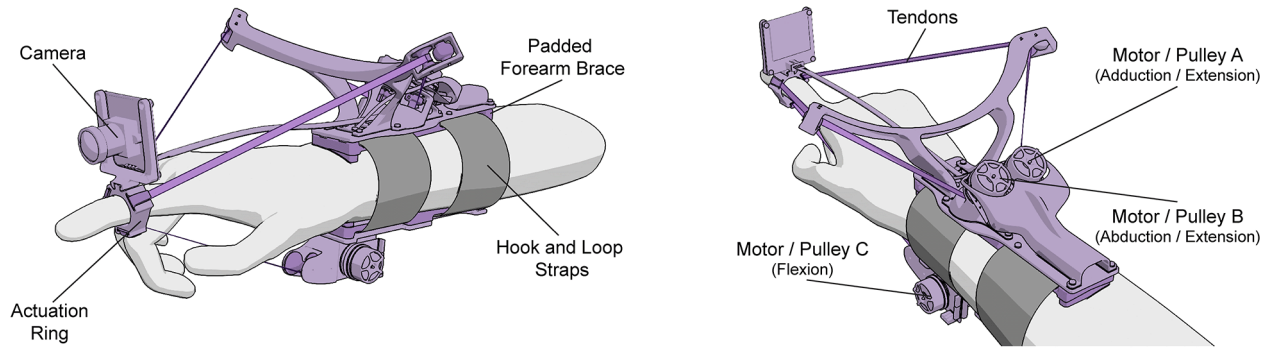


Figure 2: Components of the camera-enabled NURing prototype.

Prior work has often shifted guidance information to other sensory channels. Audio cues can be effective [10, 13, 19], but can also obscure an already critical sensory channel [15] and demand sustained attention in noisy environments [6]. Vibrotactile haptic feedback may leave hearing unobstructed, but devices worn on the hand or fingertips [2, 4, 12, 14] may interfere with natural touch and object manipulation [16]. Additionally, device-generated vibrotactile cues may be perceived as alerts rather than continuous guidance cues [5], and can require extensive training or cognitive interpretation [3, 7].

Despite decades of research into guidance technologies, it is telling that the most prevalent guidance aid for individuals with BVI remains the white cane [8]. Its continued use reflects a key principle: the cane provides continuous, direct, and naturally-understood physical feedback that allows users to feel what is happening rather than interpret symbolic cues.

Motivated by this principle, our prior work has explored fingertip deflection as a guidance cue [17, 18], inspired by the familiar act of a child leading an adult by the finger [11]. Rather than encoding commands into signals that must then be decoded, fingertip deflection gently biases the user's hand motion during reach, providing a directional cue that is continuous, immediately interpretable, and preserves user agency. This prior work [17, 18] validated fingertip deflection cues rendered by the NURing device for guiding users toward target poses in controlled 1D and 2D settings using an instrumented testbed that constrained motion to pivoting about the elbow.

To build on this foundation and explore fingertip guidance beyond controlled testbeds, we present a prototype that integrates ring-mounted vision to estimate the location of marker-tagged targets and enable closed-loop guidance during free reach. In this poster paper, we focus on demonstrating the feasibility of camera-enabled fingertip deflection guidance and providing a clear picture of the guidance behavior through trajectory and video-based demonstrations. The next section summarizes the design goals that motivate the prototype and structure our planned evaluation with BVI participants.

1.1 Contributions

- (1) We integrate a ring-mounted camera for object-referenced closed-loop deflection guidance during free reach.

- (2) We map visual target displacement to continuous guidance cues that preserve hand function and agency.
- (3) We provide two pilot demonstrations (target acquisition and object retrieval) with trajectory/video visualizations that help inform a planned BVI study.

2 Design Goals from Need-Finding

Early conversations with members of the BVI community highlighted two recurring challenges: (1) locating and safely retrieving small objects within reach, and (2) locating and operating controls in unfamiliar environments (e.g., appliances in a hotel room, controls in an elevator), increasingly mediated by flat touchscreens interfaces lacking tactile landmarks. These challenges motivated four design goals (DG) for camera-enabled fingertip deflection guidance:

- (1) **DG1—Preserve hand function:** keep the fingertips and hand available for tactile exploration and dexterous manipulation of objects and interfaces.
- (2) **DG2—Provide continuous, embodied direction:** render directional cues that can be actively followed during motion without the need to decode symbolic patterns.
- (3) **DG3—Maintain user agency:** ensure that user guidance is easy to override and supports an immediate return to self-guided exploration
- (4) **DG4—Support two-stage interaction:** provide guidance for whole-body motion during navigation / approach, followed by fine hand guidance for object manipulation.

The first two design goals have already been demonstrated in our prior work [17, 18]: applying deflection near the proximal-interphalangeal (PIP) joint keeps the fingerpad unobstructed (DG1), and rendering continuous directional cues rather than symbolic patterns enables guidance without decoding (DG2). In this work, we advance DG3 by integrating sensing directly on the wearable, enabling closed-loop guidance during free reach and building the foundation for self-guided exploration and interaction.

The ring-mounted vision system also establishes a path toward two-stage interaction (DG4): coarse guidance and environmental navigation during approach, followed by fine guidance for manipulation. The demonstrations presented here focus on peripersonal

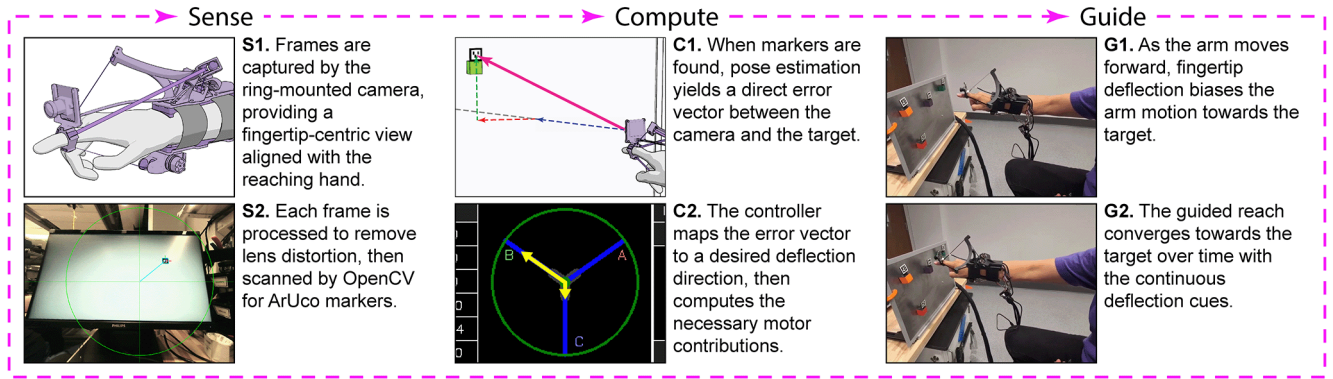


Figure 3: The camera-enabled feedback loop, used to drive continuous fingertip deflection during guidance.

fine guidance (reaching within ≈ 60 cm); broader navigation guidance over longer distances has been explored in early pilots but is not yet integrated. Future work will investigate how the same ring-mounted sensing can scale from fine contact-point guidance (demonstrated here) to room-scale navigation. In the following sections, we describe the prototype and present two pilot demonstrations that illustrate the fine guidance goals in observable behavior.

3 Prototype: Camera-Based Closed-Loop Fingertip Deflection Guidance

As the core device design has been detailed in our prior work [17, 18], here we only summarize the high-level device features before introducing the new ring-mounted camera integration that drives closed-loop fingertip deflection guidance.

3.1 Haptic Device Overview

The NURing is a tendon-driven wearable device (Fig. 2) that generates directional fingertip deflection using three independently-actuated tendons routed from a forearm-mounted drive band to a lightweight actuation ring worn on the index finger, just above the proximal-interphalangeal (PIP) joint. Coordinated tendon actuation produces deflection cues across the anatomical directions (abduction, adduction, flexion, and extension), allowing the system to continuously bias the user’s hand toward a given direction during the reaching motion. This implements DG2 by providing continuous directional bias during motion, while applying forces off of the fingertip to preserve touch-based interaction (DG1). The prototype device presented here, with the inclusion of the camera, weighs approximately 300 grams, similar to a smartphone.

3.2 Computer Vision Integration

In previous studies using the NURing [17, 18], closed-loop guidance relied on external sensing of forearm orientation measured within an instrumented platform. New in this work is the integration of a wide-angle global-shutter camera (120-degree field of view), mounted on the actuation ring to provide a fingertip-centric view aligned with the guided hand. This new sensing modality enables fingertip-centric localization during free reach and provides a sensing pathway toward interaction discovery (DG4). The

camera captures 1600x1200-pixel frames at 60 Hz, enabling responsive closed-loop guidance during reach. Frames are processed and undistorted using intrinsic calibration to improve pose estimation.

We use OpenCV to detect fiducial markers present in the camera frame using the ArUco library. For each detected ArUco marker (Fig. 3, S1-S2), the system estimates the marker’s 3D translation in physical units relative to the camera, resulting in a target position in the camera’s coordinate frame (Fig. 3, C1). Because the camera is mounted to the ring and therefore co-located with the finger, the measured displacement between the target and the camera’s principal point provides a direct error vector for closed-loop guidance. At each frame, the system computes this error vector and then maps the vector heading to a deflection direction. The proportional controller then scales the magnitude of this deflection and computes the coordinated motor contributions needed to impart fingertip deflection in the direction of the target (Fig. 3, C2). This produces a continuous embodied guidance signal (DG2) that biases the user’s ongoing arm motion (Fig. 3, G1-G2) while remaining easy to override (DG3), allowing the user to converge on the object through continuous, direct embodied guidance rather than symbolic interpretation.

When tracking is lost, the system can optionally render a brief pull-and-release pattern on the three tendons together to signal uncertainty and encourage reorientation. This cue serves as a safety/uncertainty mechanism rather than directional guidance, and the intensity of such a cue as well as how it is perceived will be explored in future work to preserve comfort and user agency.

While the ArUco markers used in these pilot demonstrations provide a robust and repeatable target representation for prototyping, the guidance control pipeline itself is independent of the localization method. The system requires only a target position estimate relative to the camera frame as an input, which can be derived from marker detection, object recognition, depth sensing, or virtual coordinates in mixed-reality applications. The core elements that enable intuitive guidance remain unchanged in markerless tracking, and while transitioning to tracking everyday objects introduces challenges (e.g., detection reliability, occlusion handling) common to all vision-based guidance systems, these challenges represent active areas of computer vision research rather than limitations specific to fingertip deflection as a guidance modality.

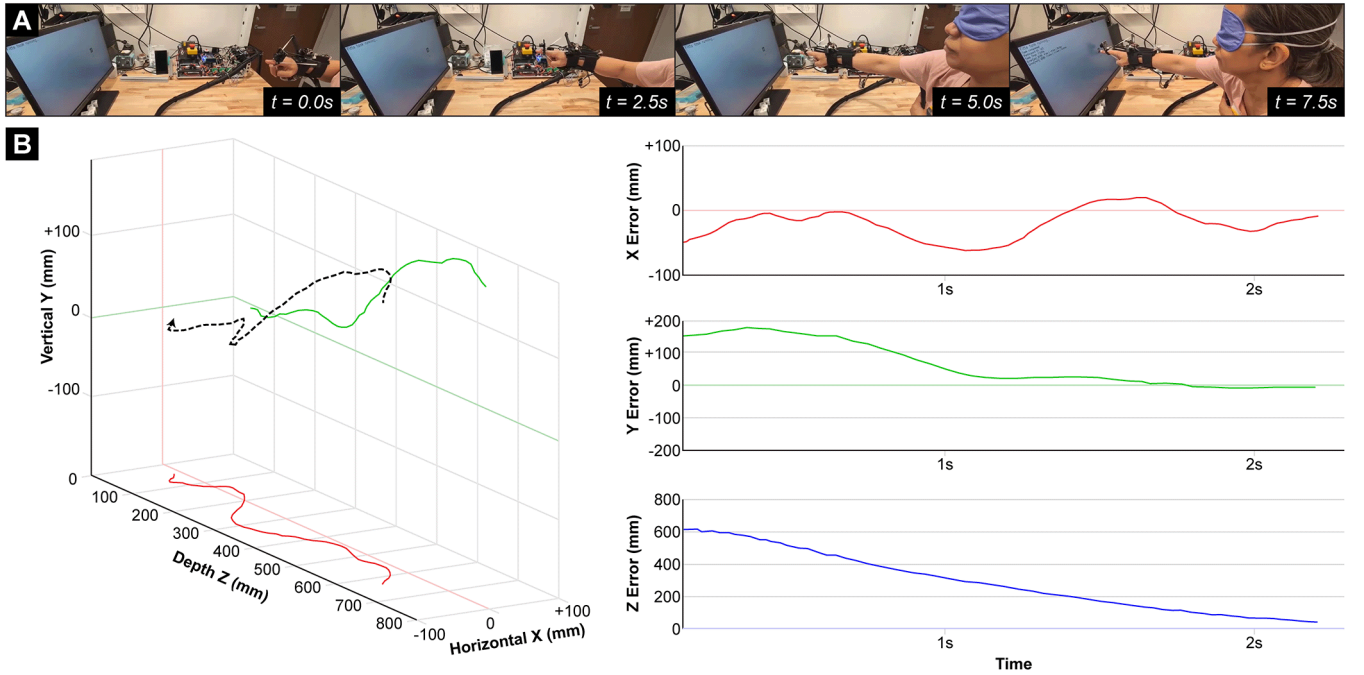


Figure 4: Target Acquisition Task: (A) Extracted video frames showing continuous guidance towards the target ArUco marker on the touchscreen; (B) Plots of an example measured trajectory; the plot at left shows the 3D trajectory in black, with color-coded "shadows" to the right, showing the shaping of the trajectory towards the target.

4 Pilot Demonstrations

We present two pilot demonstrations that illustrate camera-enabled fingertip deflection guidance for peripersonal interactions. These demonstrations are not intended as a controlled evaluation; rather, they provide feasibility evidence and visual intuition for how continuous deflection cues shape reach trajectories and support object-directed interaction. The tasks and accompanying visualizations also serve as templates for a forthcoming study design with BVI participants. Lastly, these demonstrations illustrate the current prototype capabilities under fiducial-based target localization; automated interaction discovery in unstructured environments is not yet integrated.

4.1 Target-Acquisition Task

The first demonstration task is a Fitts-style target acquisition task used to visualize guidance and trajectory dynamics. In this task, a participant is asked to move their arm forward towards a touchscreen approximately 600 mm away, upon which a 20-mm target ArUco marker has been displayed (Fig. 4A). During the task, the participant wears a blindfold to remove visual guidance cues. As they advance their arm towards the touchscreen, the NURing's continuous fingertip deflection cue generates a bias in the arm movement toward the marker-tagged target on the touchscreen. When the touchscreen detects fingertip contact, the final position and time taken to complete the task are recorded, as well as the trajectory taken during the task (converted into the real-world coordinate frame with a static target and moving camera). Figure 4B shows

an exemplar set of plots of the recorded trajectory during a pilot study (with a different participant than A). These pilot trajectories illustrate how the deflection cues shape the trajectory of the arm during reach as it travels approximately 60 cm towards the target, resulting in a final fingertip deviation of approximately 10 mm from the target center in under 3 seconds.

This task isolates the closed-loop "direction-following" behavior and provides a controlled visualization of how the cue continuously shapes the reach trajectory, supporting our goals of embodied guidance without symbolic, cognitive translation (DG2). In future work, we will use this task to quantify how performance is impacted by distance, controller tuning, and cognitive loading. A current limitation of this demonstration is the use of fiducial targets on a display; extending this to real-world targets will require more robust and refined target interaction identification and pose estimation, as well as developing strategies to handle uncertainty (as discussed in Sec. 3.2).

4.2 Object-Retrieval Task

The second task expands on the previous task to include an object-retrieval component. As in the previous task, a participant is asked to move their arm forward following guidance cues, however, this time the targets are physical objects: five 25-mm cubes magnetically-affixed to a target board. Participants are tasked with collecting each object in a randomly-generated, predefined order, and depositing them in a tray at their lap. Collecting all five target objects in the correct order represents successful task completion. As before, the participant wears a blindfold to remove visual guidance cues.

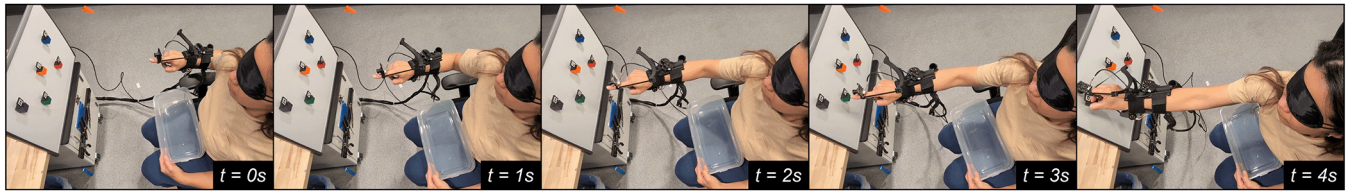


Figure 5: Object Retrieval Task: Extracted video frames showing continuous guidance toward a physical target over a four-second trajectory, followed by grasp.

Figure 5 presents an exemplar sequence of video frames from a pilot study, showing successful retrieval of a single object in around 4 seconds. The participant presented in these video frames was considered a novice user (i.e., < 2 hours of device use), and in this brief pilot session, they retrieved all five objects in approximately 40 seconds. This duration includes two reach attempts in which tracking was lost, and the participant experienced the “re-acquire” cue (as described in Sec. 3.2), leading them to pull their hand back to re-acquire the target and resume object retrieval. The planned BVI study will further evaluate user responses to tracking loss / uncertainty cues.

This demonstration extends target acquisition to a manipulation task, emphasizing that fingertip deflection guidance can be provided while leaving the fingertip pad unencumbered for touch and grasp (DG1). The re-acquisition behavior also highlights an important practical requirement: guidance systems must fail safely and support quick recovery without removing user control (DG3). A limitation of the current prototype is sensitivity to tracking loss during occlusion, for which we are developing models to predict target object pose based on last-known readings.

4.2.1 Scope of pilot demonstrations. Together, these pilots demonstrate (1) the technical feasibility of ring-mounted vision for closed-loop guidance during free reach, and (2) visual intuition for how deflection cues shape trajectories and support object interaction. At the same time, they expose key design questions around cue semantics near contact and robust recovery under tracking loss. They do not provide systematic performance evaluation, comparative analysis against other modalities, or generalize usability findings. A planned study with BVI participants will address these questions through controlled evaluation of task performance, workload, and perceived agency.

5 Conclusion

In this paper, we extend our previous work [17, 18] beyond instrumented testbeds by integrating a camera mounted on the actuation ring, enabling real-time target localization from marker-tagged objects for closed-loop guidance during reach. We illustrate the resulting camera-enabled interaction through two pilot demonstrations: (1) a Fitts-style target acquisition task and (2) an object-retrieval task. These demonstrations provide feasibility evidence and visual intuition for how continuous deflection cues can shape reach trajectories while preserving hand function and user agency. Building on these demonstration tasks, we are preparing a study with BVI participants to evaluate task performance (success rate, completion time, path efficiency) and subjective user experience (NASA-TLX

workload, perceived agency, confidence ratings, comfort), among other metrics. This controlled evaluation with the target user population will validate the NURing’s fingertip deflection guidance for assistive contexts and identify opportunities for refinement. By shaping arm motion through direct, embodied cues without relying on vision or occupying other sensory channels, this work highlights camera-enabled fingertip deflection as a practical guidance modality for real-world interaction, supporting future assistive and collaborative systems that guide action through intuition rather than symbolic, cognitive translation.

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